



How do I do DevOps when all I have is Ops?

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DevOps Borat

@DEVOPS_BORAT

Confuse of Dev or Ops? Simple rule: if you are praise for Web site success, you are Dev; if you are blame when Web site down, you are Ops.

Agenda

- Organisational Context
- What do I mean by DevOps?
- All in on Operational Data Mining
- People, Process and Tools



The organisational context



Let's get Conway's law out of the way early

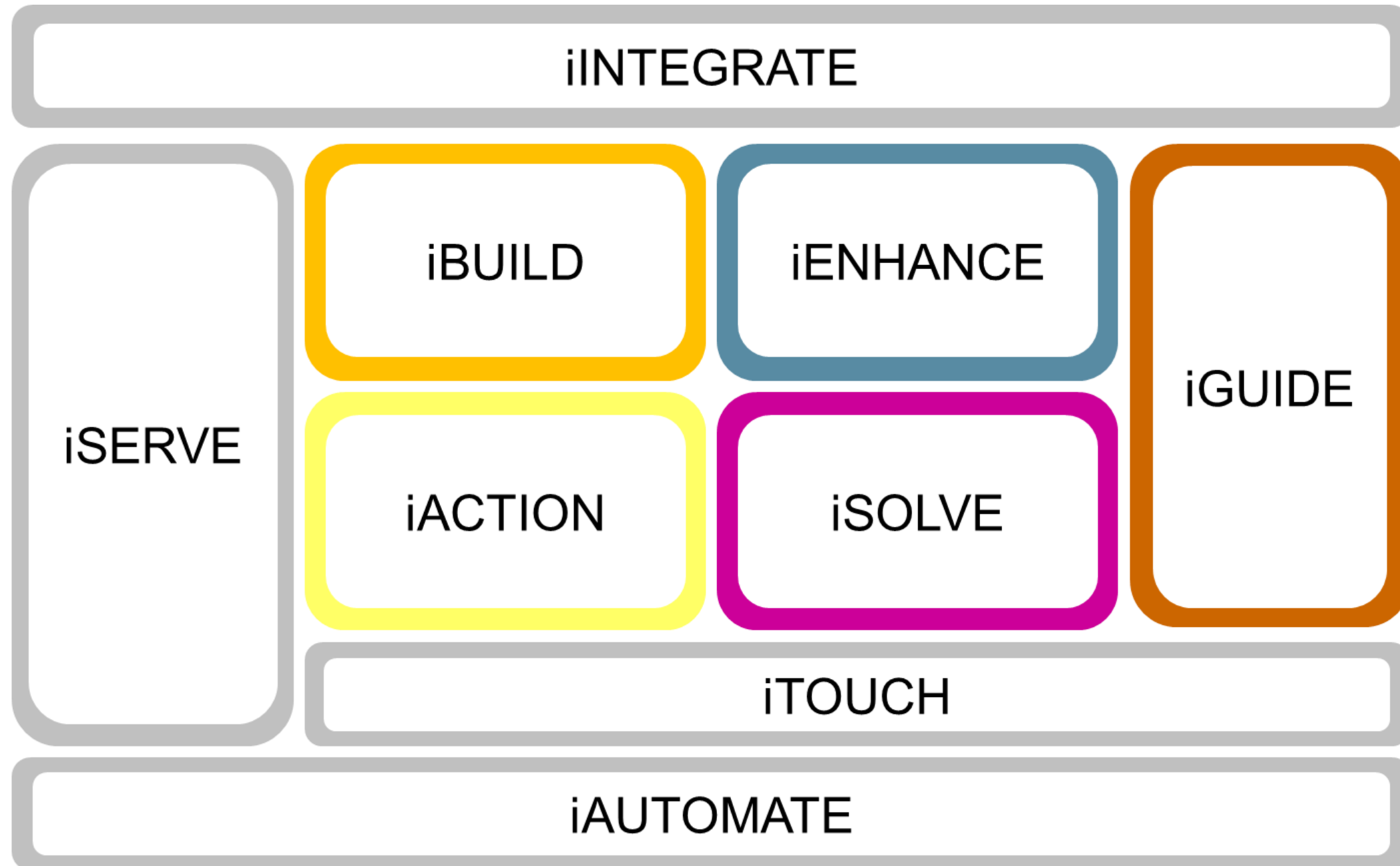
Dev

Ops

GBS

GIS

iDo – how we presently organise delivery





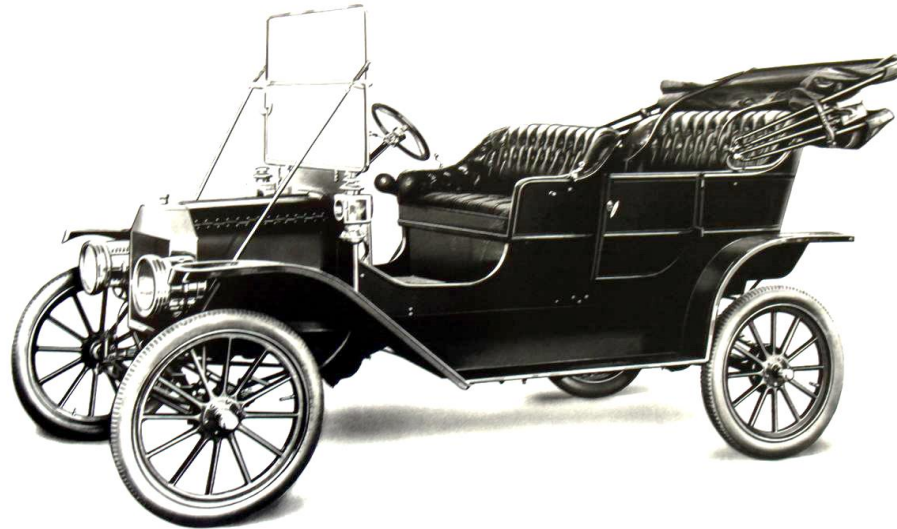
What do I mean by DevOps?



All industries evolve



Design for Purpose

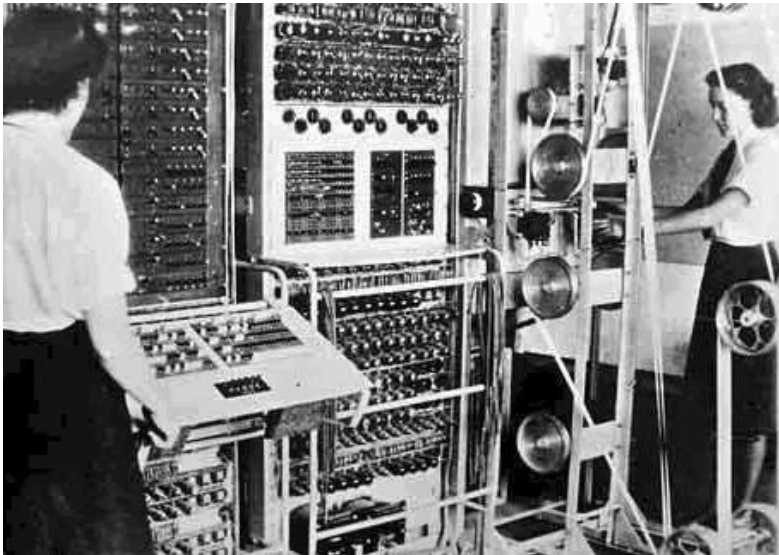


Design for Manufacture



Design for Operations

IT is no different



Design for Purpose



Design for Manufacture



Design for Operations

Agility, Agility, Agility

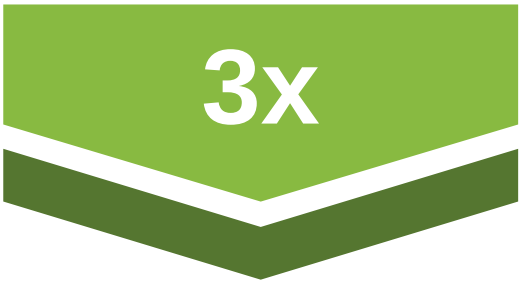
High-performing IT organizations report experiencing:



more frequent deployments



faster recovery from failures

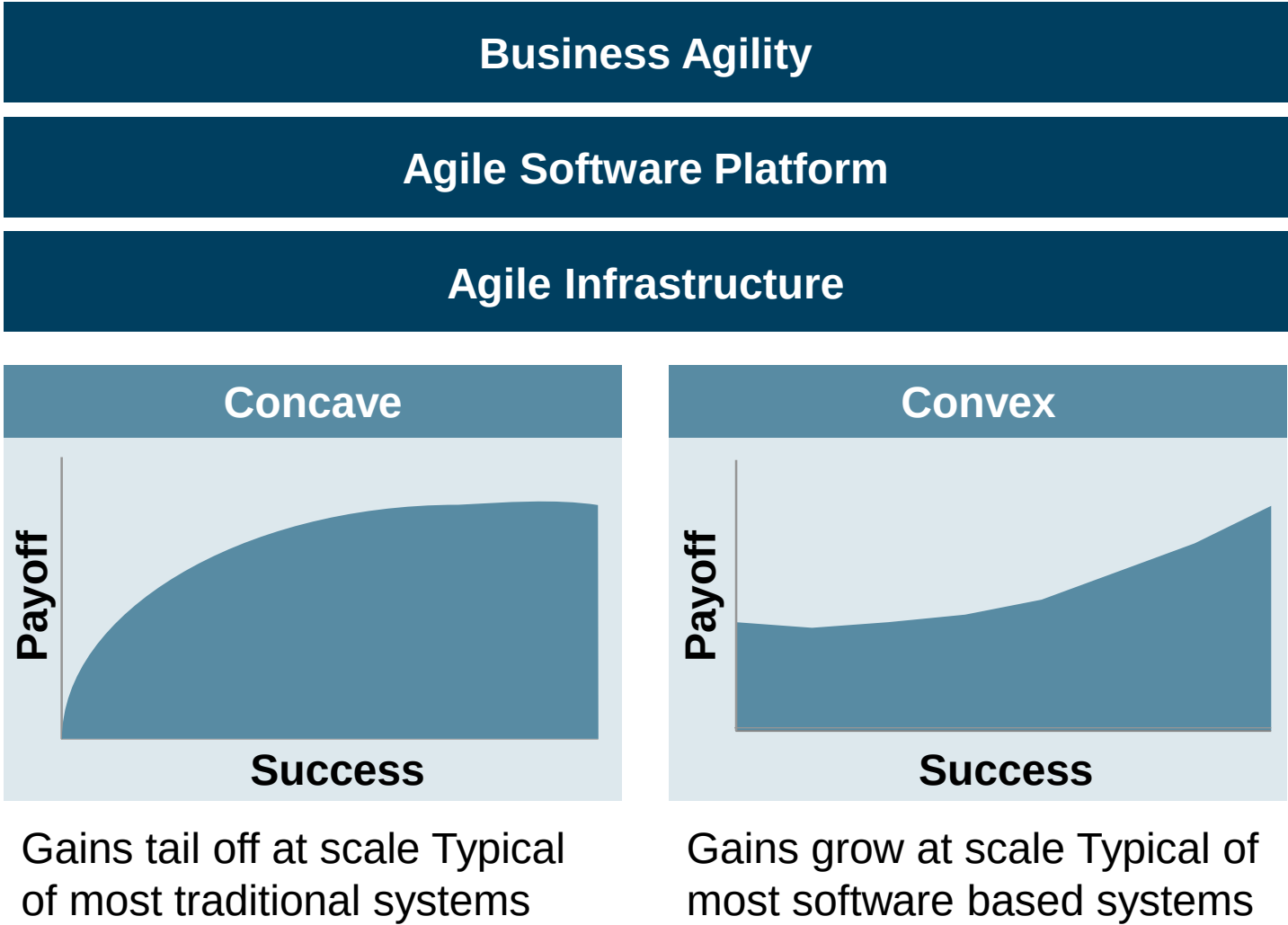


lower change failure rate

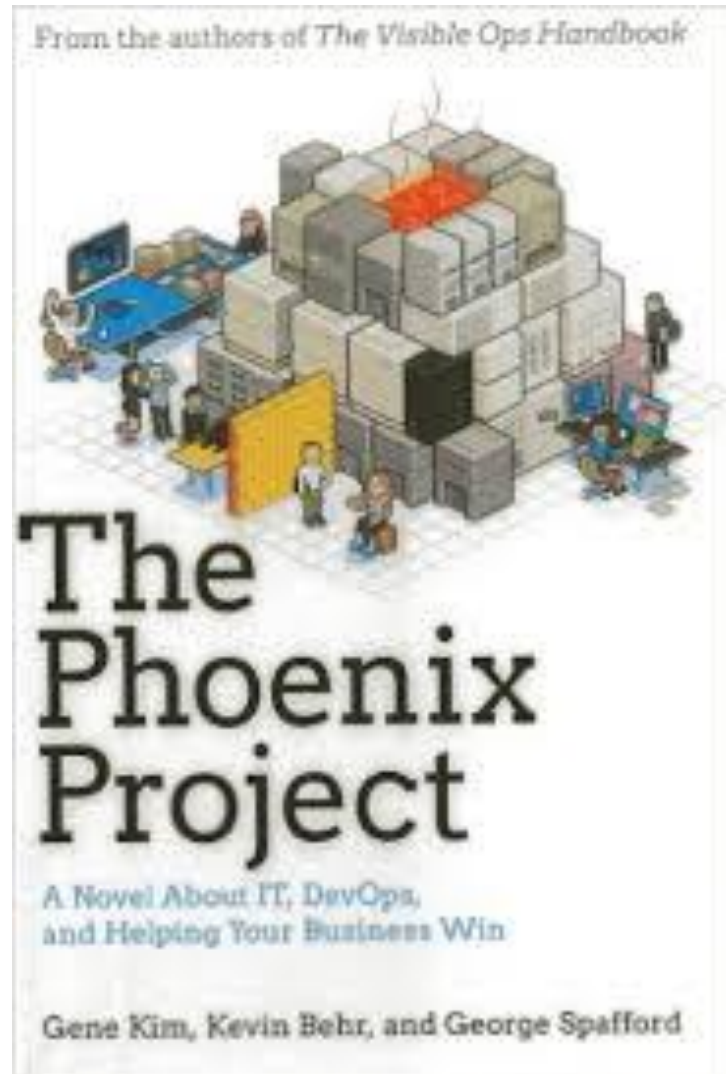


shorter lead times

Source: 2016 State of DevOps Report



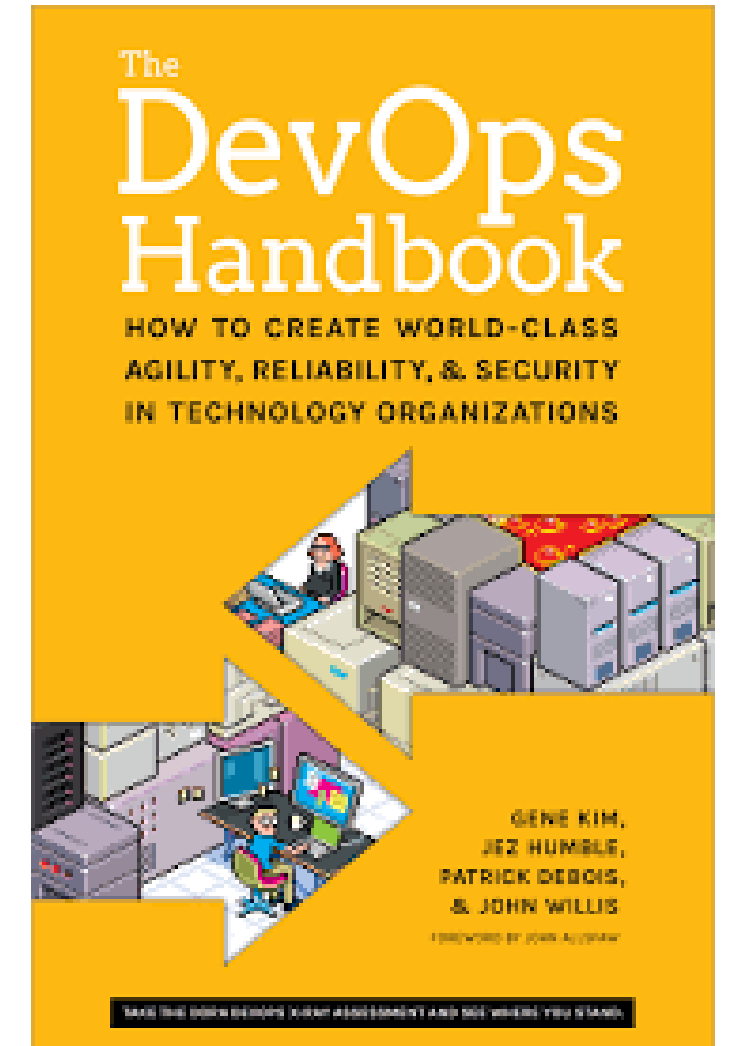
The 3 ways



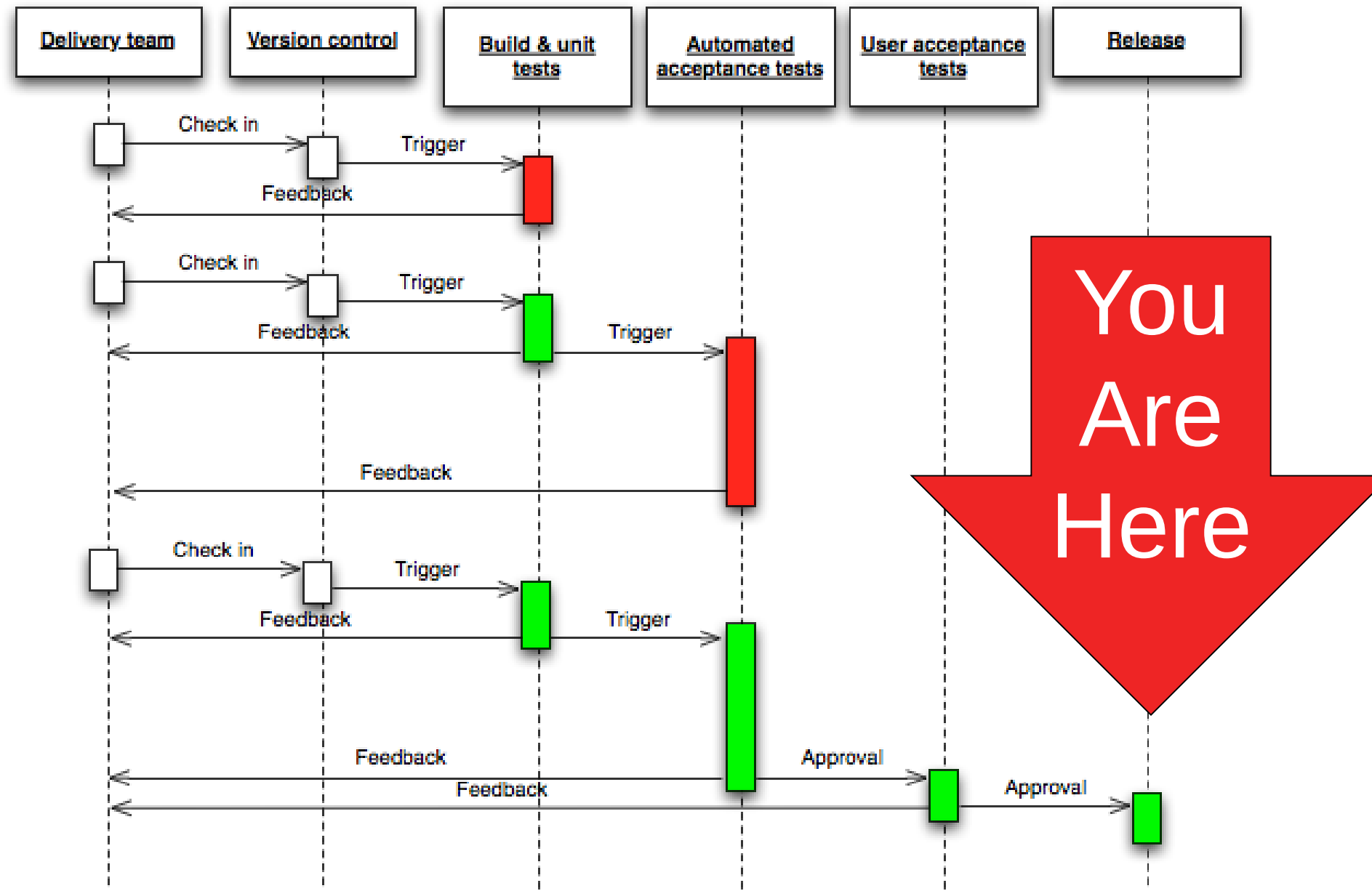
1.Flow

2.Feedback

3.Continual Learning
& Experimentation



Flow – people want Continuous Delivery, and we can help with that, but it's not something that can be assumed or directly influenced



Feedback – it’s pretty much the same story as with Flow

 **Build #340 [b61e6d09] (Jun 29, 2015 2:11:59 AM)**

Started 10 hr ago
Took [5 hr 49 min](#) on master

 [add description](#)



Changes

- 1. Fix mistake introduced with support for gcc 5.1. ([detail](#))

ations in the build



Started by timer



Revision: b61e6d0991e8eae717997c754d9b52594c513a11

- origin/v5-34-00-patches



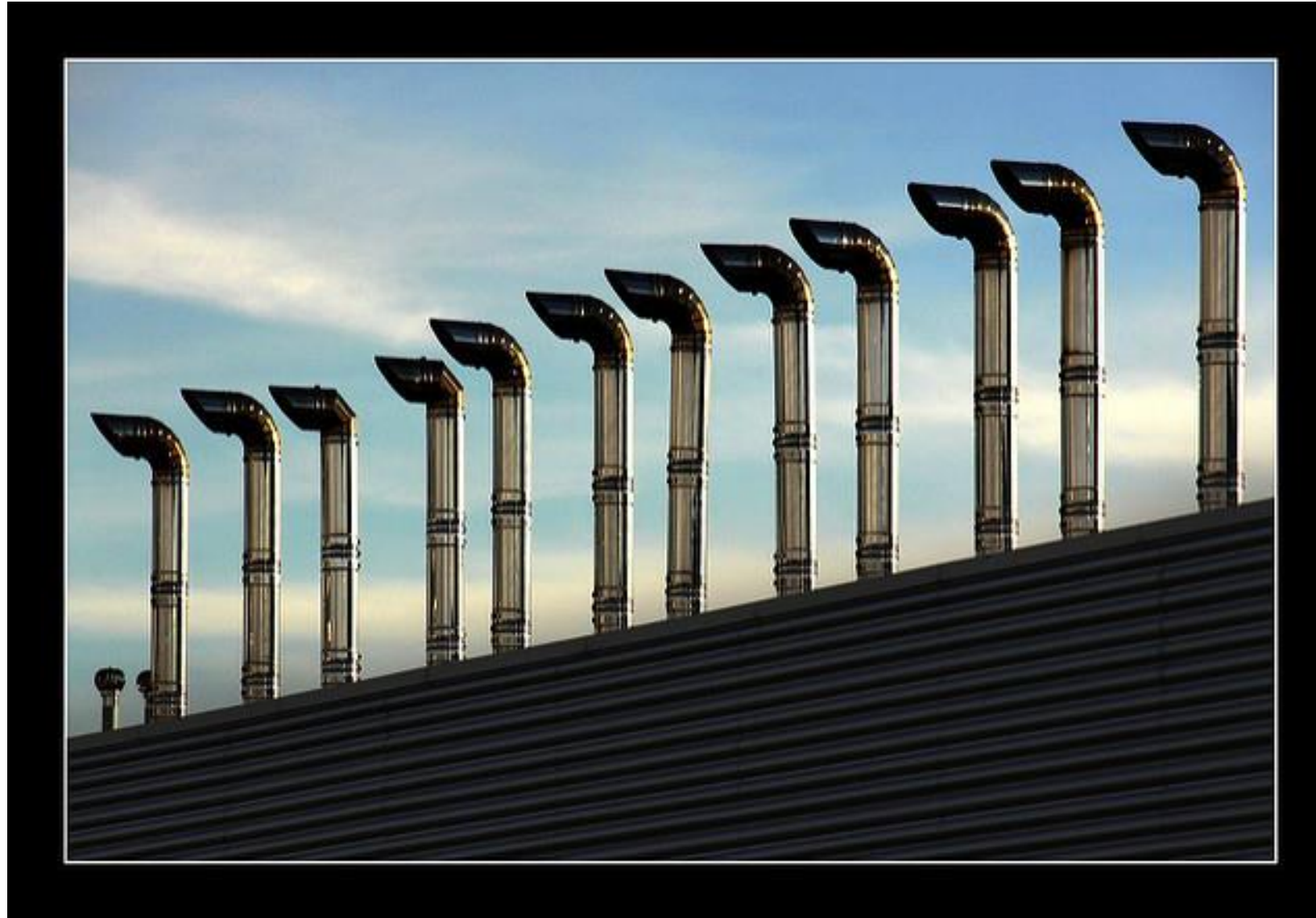
GNU C Compiler Warnings: [1 warning](#).

- [12 fixed warnings](#)

Configuration Matrix		gcc48	gcc47	gcc49	gcc51	icc	clang34	clang36	native	vc9	vc10	vc11	vc12	classic
cc7	Debug													
	Release													
fedora20	Debug													
	Release													
mac1010	Debug													
	Release													
mac108	Debug													
	Release													
mac109	Debug													
	Release													



Operational Data Mining (ODM) provides a means for Continual Learning and Experimentation



'Exhausting' by Ben Salter <https://flic.kr/p/8VTaMe>



All in on ODM



Data helps us find the constraints, then tells us what to do with them



'Narrow' by gwire <https://flic.kr/p/4d3N4>

Data provides a means of empowerment to front line staff

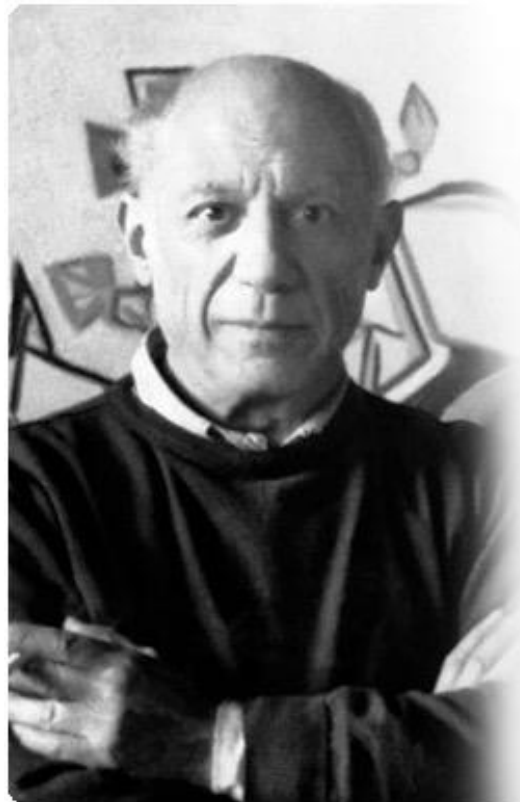
*“I knew that,
I knew that we needed to do that”*



People, Process and Tools



Infrastructure as Code boot camp



"Good artists copy,
great artists steal."
- Pablo Picasso

vantucker.me

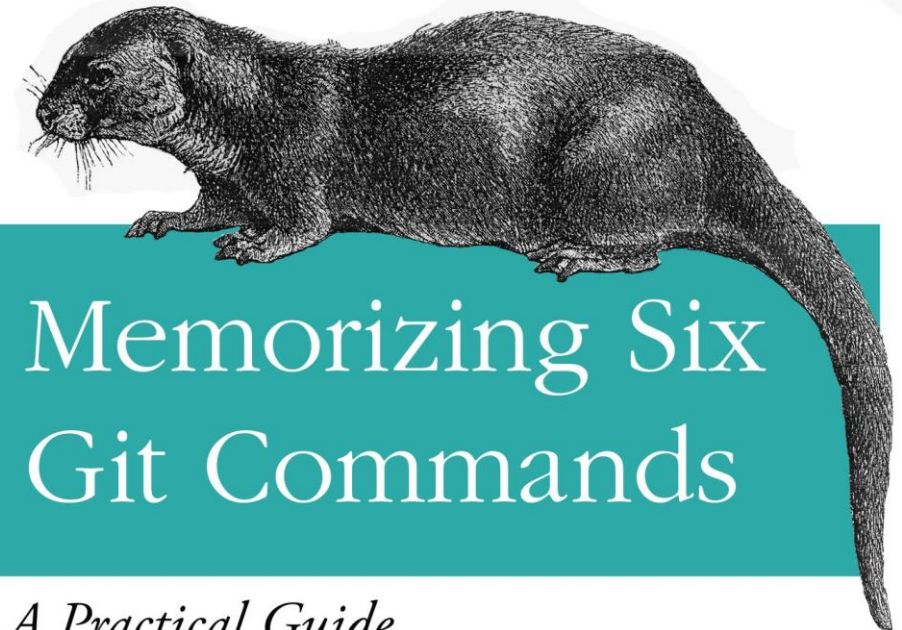
Infrastructure as Code boot camp - Git

Git Goals

- Fork a repo
- Create and manage SSH keys
- Clone the fork
- Modify the fork by adding a file with short name
- Add and commit a file to git
- Push the change to GitHub
- Make a pull request

Part 1 Git and GitHub

The popular approach to version control



A Practical Guide

O RLY?

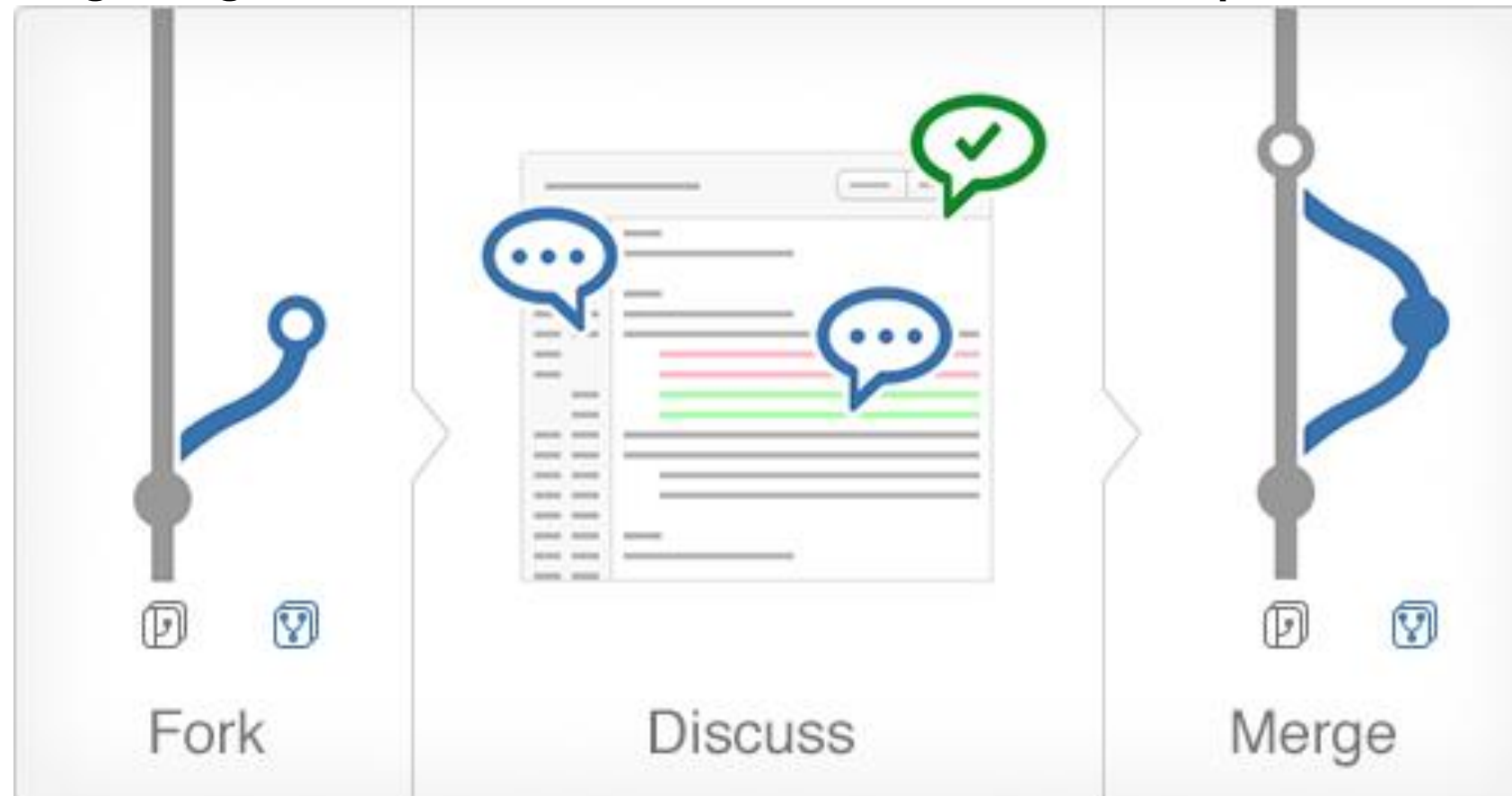
@ThePracticalDev



We pay particular attention to Fork'n'Pull

Beg forgiveness

Ask permission

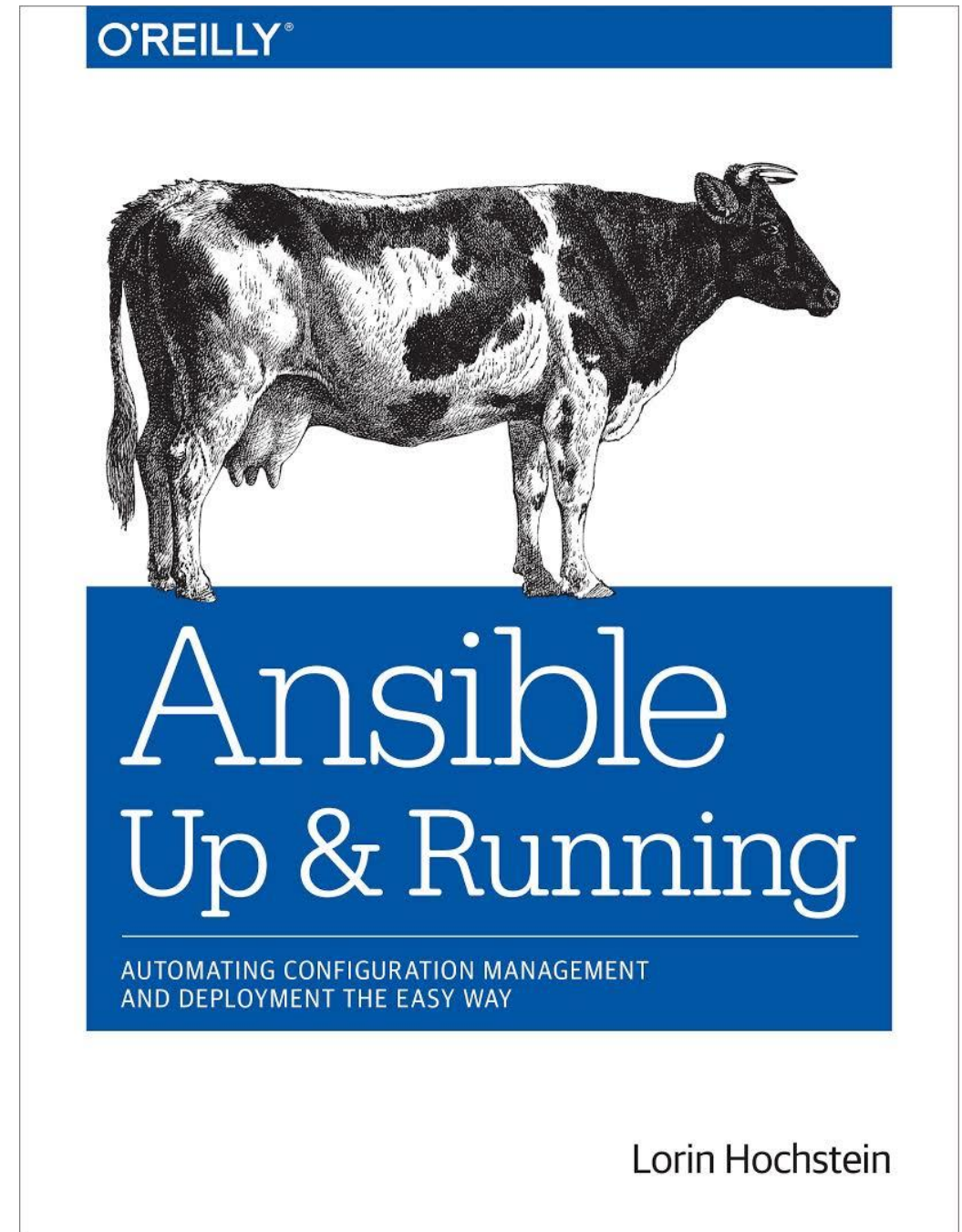


Infrastructure as Code boot camp - Ansible

Ansible Goals

- Clone an existing repo containing playbooks
- Set up SSH key based authentication
- Ad hoc Ansible commands
- Use a playbook to install a web server
- Use Ansible to change the web server config

Part 2 Ansible

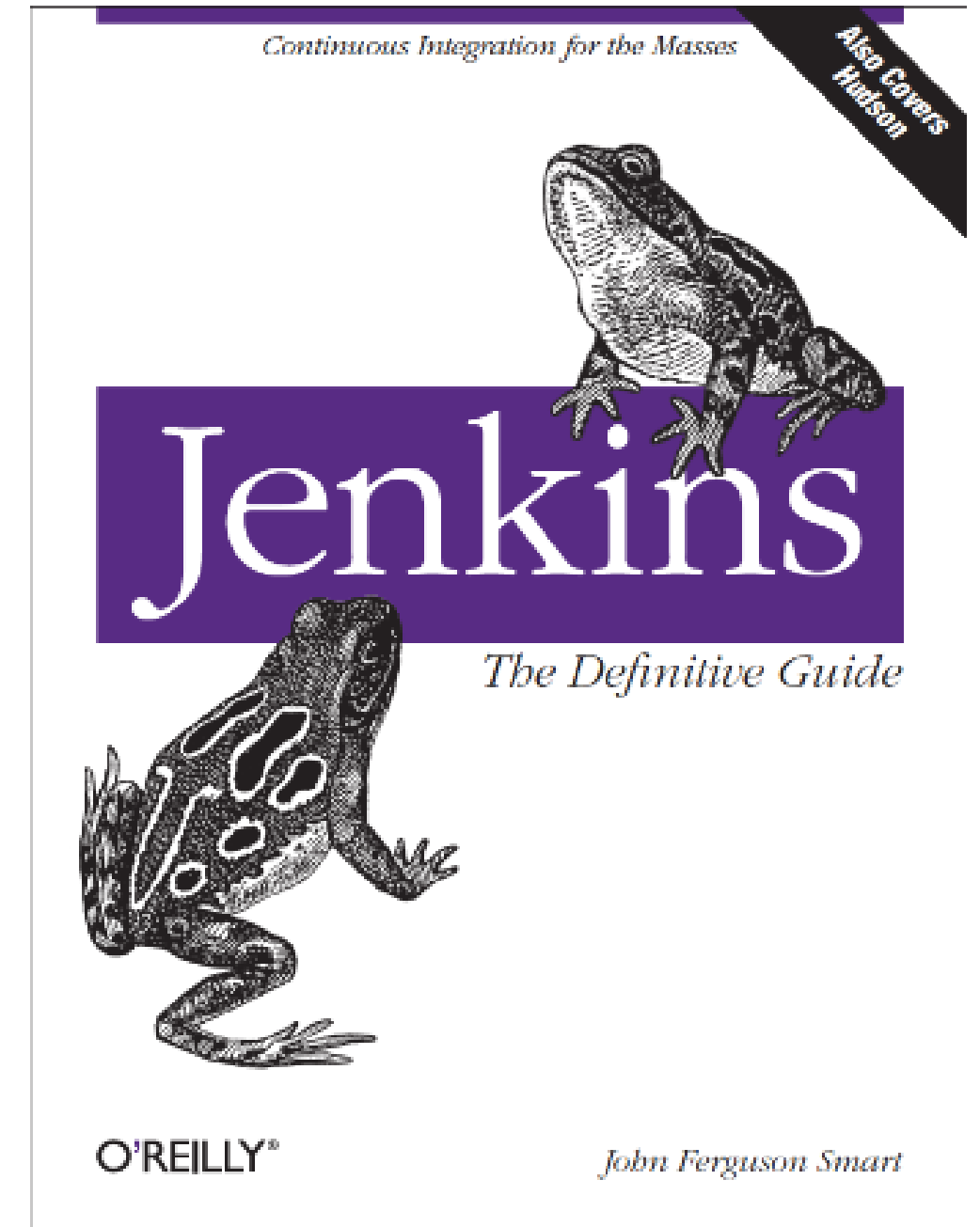


Infrastructure as Code boot camp - Jenkins

Jenkins Goals

- Configure Jenkins
- Build an example project
- Set up an app server to deploy to
- Deploy to the app server from Jenkins

Part 3 Jenkins



The perfect runbook...



'Snug' by Thomas <https://flic.kr/p/qbdT3c>

Doing root cause analysis properly

Why?

Why?

Why?

Why?

Why?



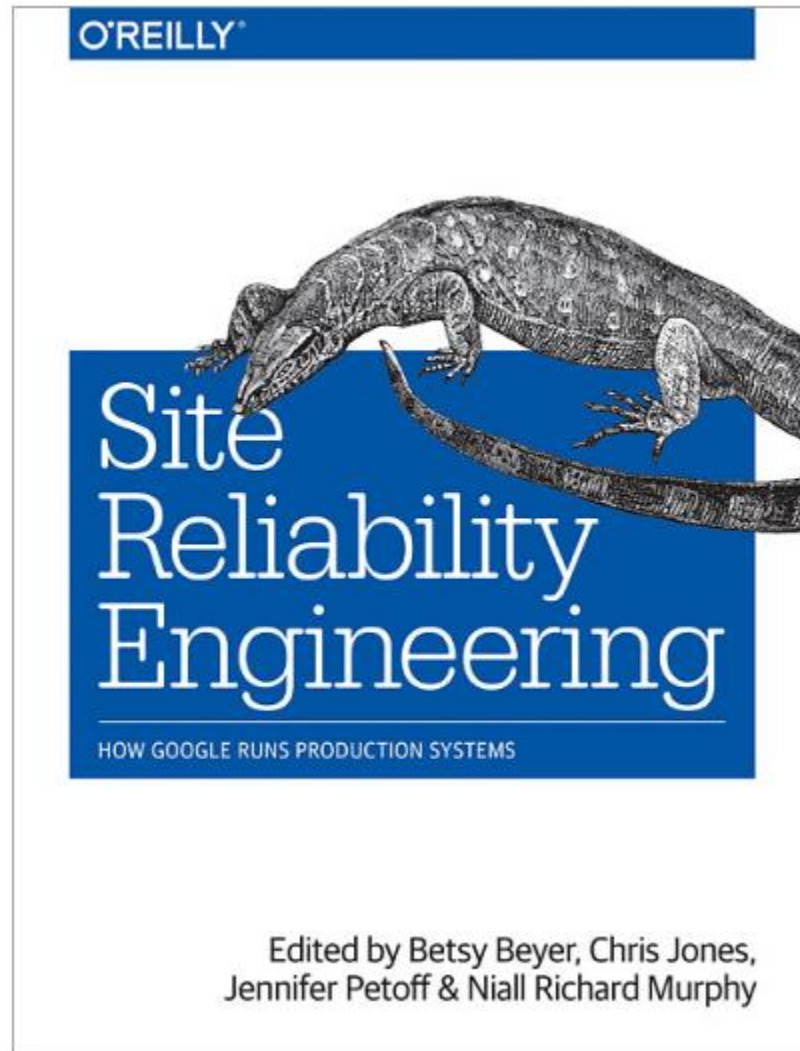
Wrapping up



Review

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- What do I mean by DevOps?
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Further reading



Final Thought



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To make error is human. To propagate error to all server in automatic way is #devops.



Questions?

